

Numerical Methods Lab 02 - Results

Palaganas

Fitted Parameters

Parameter	Value	Std. Error	t-statistic	p-value
c	5.008684	0.006216	805.7887	0.0000
a	2.992357	0.008290	360.9805	0.0000
k	0.819642	0.009484	86.4238	0.0000
t0	10.026408	0.016148	620.8880	0.0000

Model Statistics

SSE (Sum of Squared Errors)	0.080036
R ² (Coefficient of Determination)	0.999518
s (Standard Error of Estimate)	0.029495 meters
SST (Total Sum of Squares)	166.195591
n (Number of data points)	96
p (Number of parameters)	4
Degrees of Freedom	92

Area Under the Curve

Area from fitted curve: 160.0212 meter-hours

Trapezoid cross-check (raw data): 160.0187 meter-hours

Integration limits: t = 0.00 to t = 23.75 hours

Reading of Residuals

The residuals appear to be randomly scattered around zero, with no obvious trend or pattern. This suggests that the logistic model provides a good fit to the data. The magnitude of the residuals is small (within ± 0.1 meters), indicating that the model captures the overall behavior of the reservoir level well. There are no significant outliers or systematic deviations, which confirms that the chosen model is appropriate for this dataset.